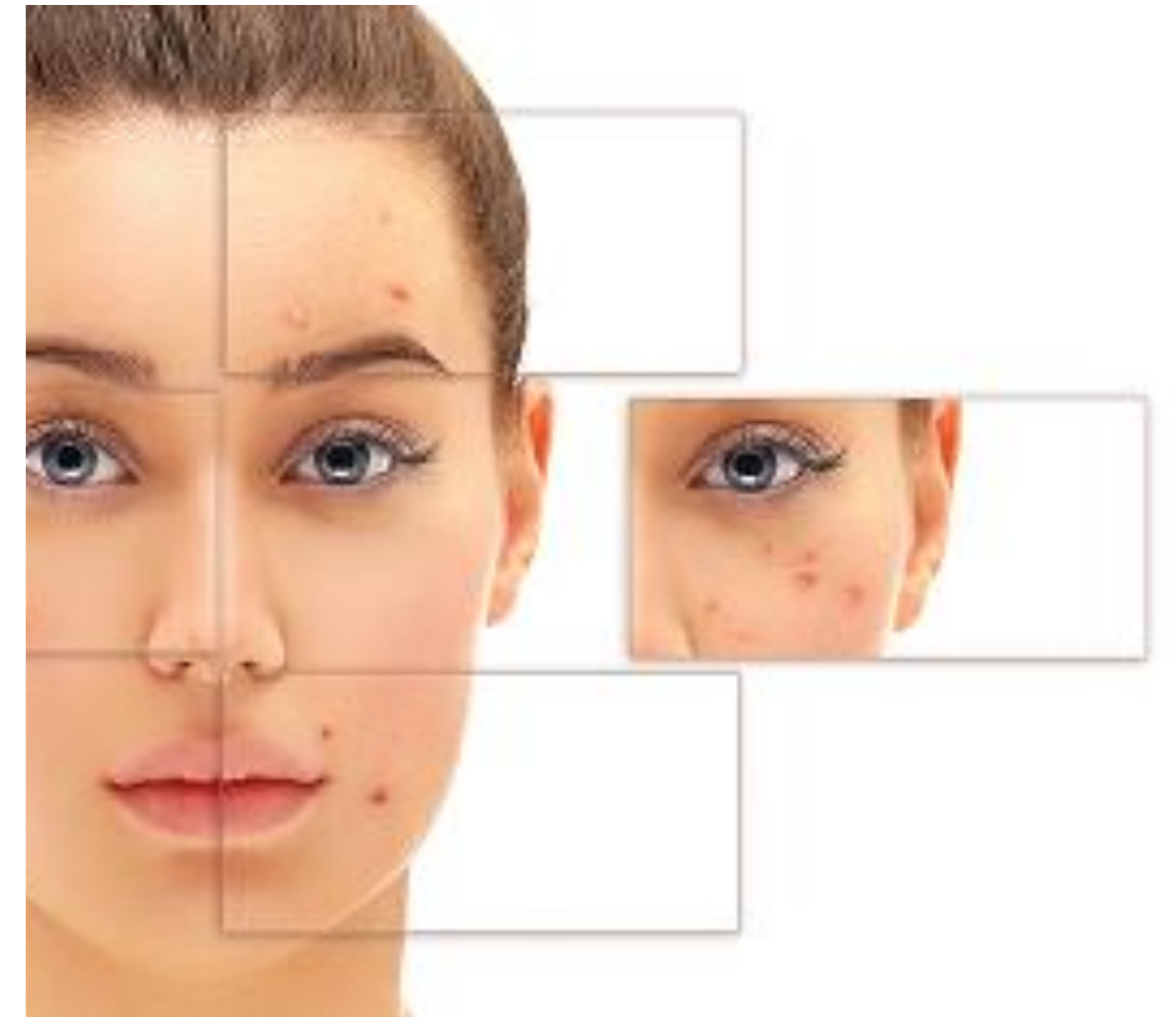


Multimodal AI Framework for Personalized and Culturally Aware Skin Disease Diagnosis, Monitoring, and Treatment Support.



GROUP : 25-26J-222



GROUP MEMEBERS



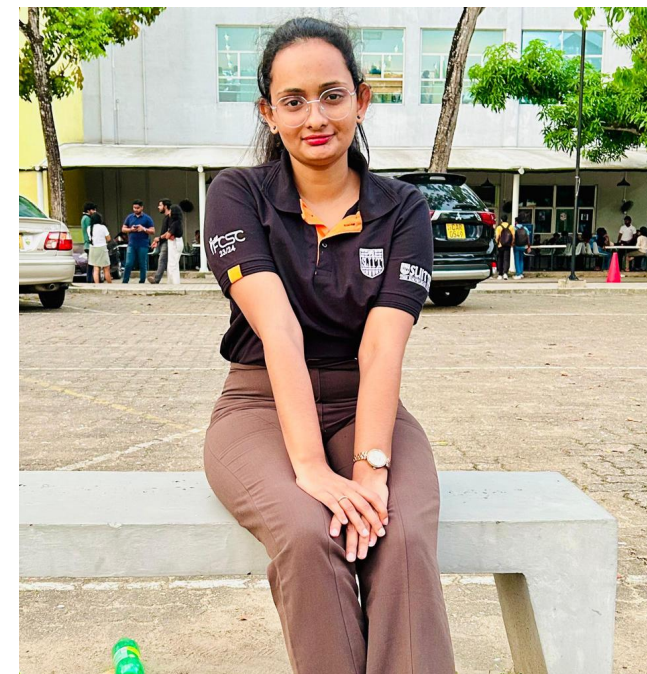
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INTRODUCTION

This research proposes an **AI-powered skin care support platform** that analyzes patient symptoms and optional skin images using advanced machine learning models while supporting multilingual interaction.

The system aims to improve early detection, provide personalized guidance, and enable continuous, fair, and accessible skin disease management for teenagers and adults

Problem Statement & Stakeholders

- Context:** Rural and underserved areas with limited access to dermatologists

.

- Stakeholders:** * **Primary:** Teenagers and adults with skin concerns.

Secondary: Healthcare providers and caregivers.

- Observed Issue:** Existing AI solutions have bias in skin tone detection and lack continuous monitoring/personalized guidance.

- Impact:** Leads to unsafe self-medication and delayed treatment for darker skin tones.

RESEARCH GAP AND OBJECTIVES

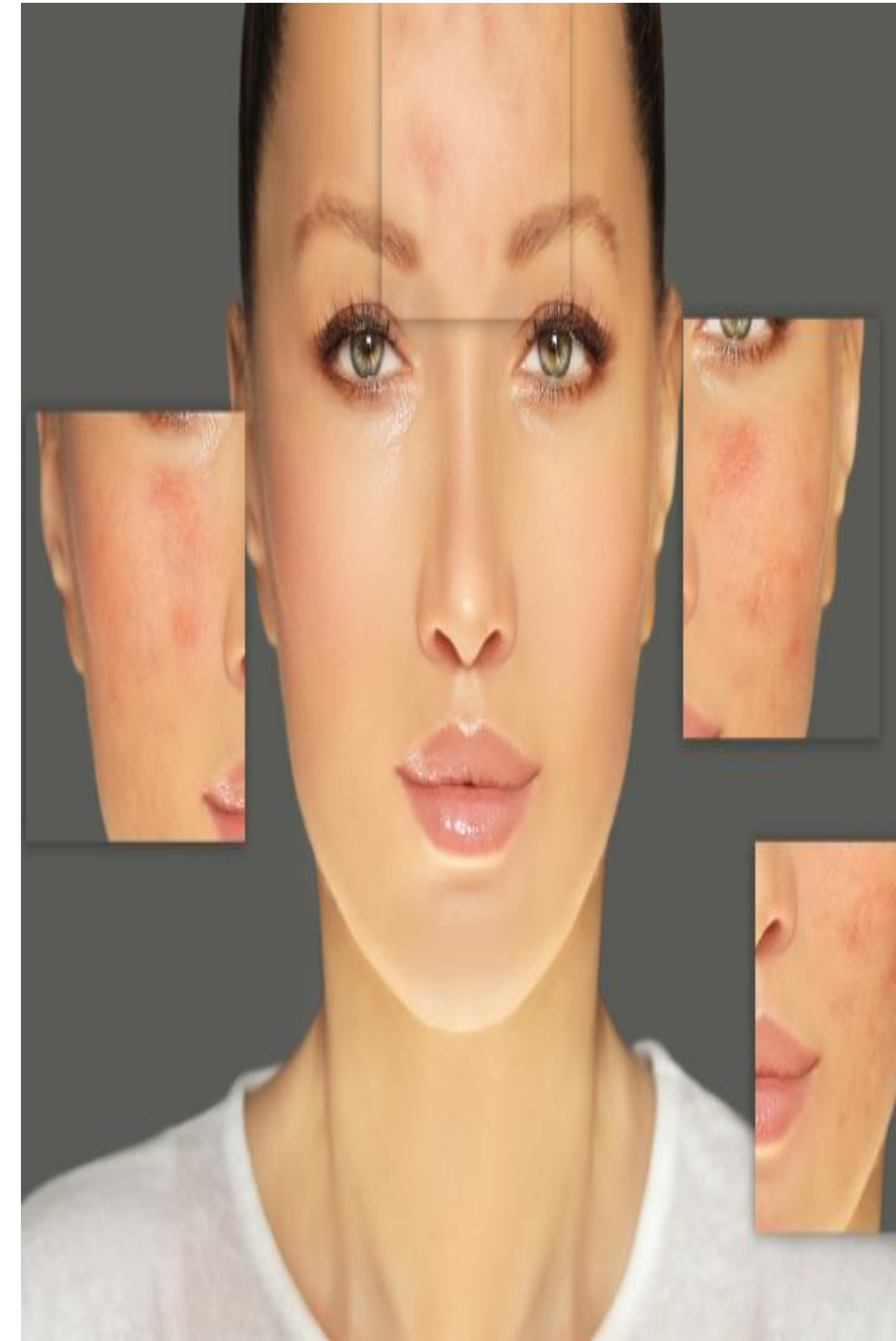
RESEARCH GAP

Most apps are "one-shot" diagnosis tools that fail on diverse skin tones and don't provide long-term tracking or multilingual support.

Most app hasn't aimed the audio-base diagnosis and not mentioned the treatment supports.

OBJECTIVES

To develop a fair, multimodal AI framework that provides skin disease diagnosis, monitoring, and multilingual treatment support with 90% accuracy across skin tones



Solution Overview

- Summary:** A comprehensive AI-powered skin care platform focusing on identify diagnosis and continuous patient support.

- Key Features (FRs):**

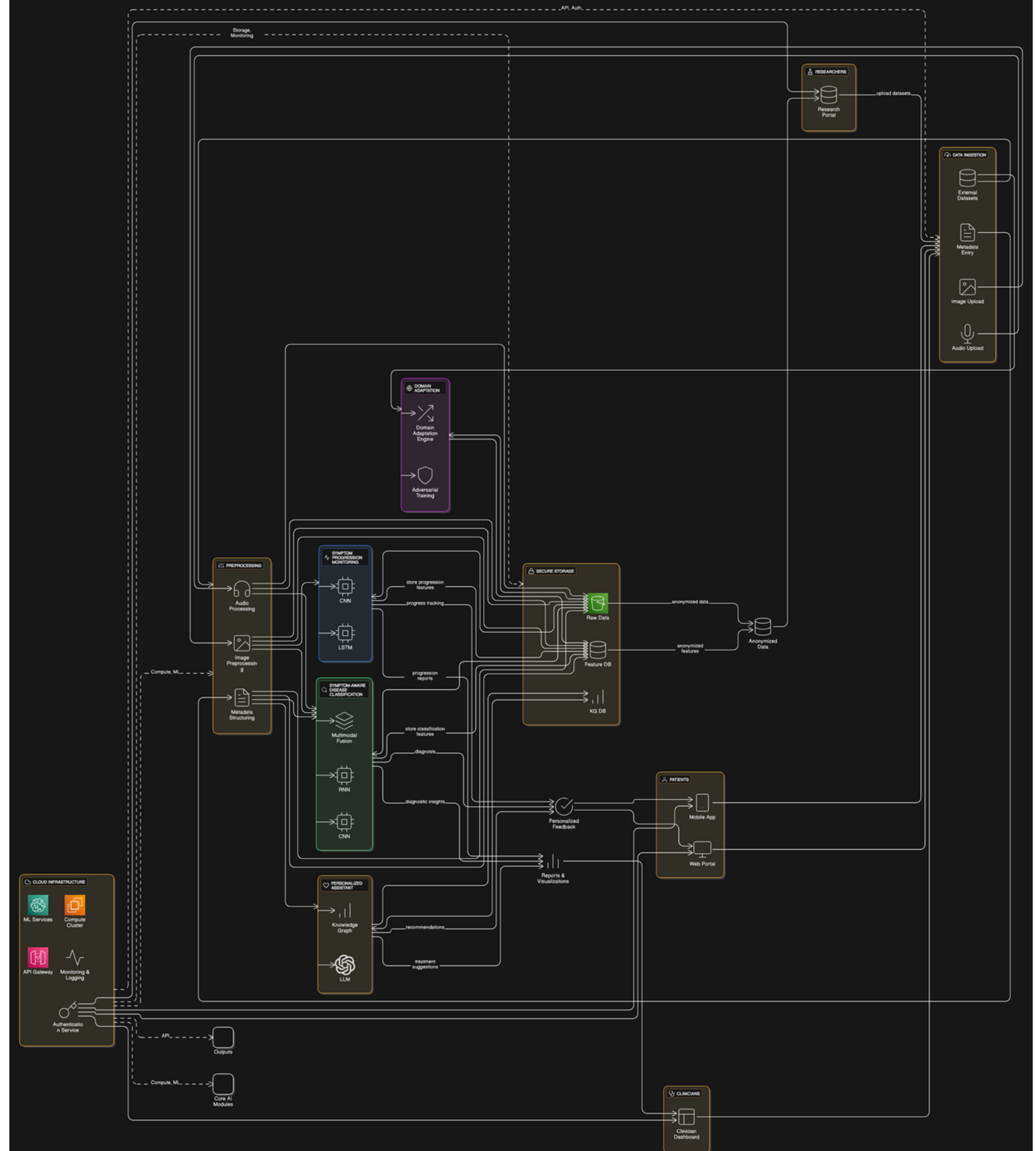
1. Skin care supporting
2. image and audio base diagnosis clarify
3. Text base diagnosis clarify and treatment support.
4. Monitoring the condition of diseases.

- Status:** ~90% Complete



SYSTEM OVERALL DIAGRAM

- **Frontend:** Mobile Interface(Flutter).
- **Backend:** FastAPI .
- **AI Engines:** CNNs for image classification + NLP models for symptom analysis.
- **Database:** Patient history and tracking logs. Use super base database.
- **Integration:** APIs for model serving; JSON data formats for cross-component communication.



Commercialization

Investment & Cost Recovery

- **Investment/Cost:**
 - *Labor:* AI/ML engineers, developers, domain experts.
 - *Material:* Cloud hosting, storage, APIs, security & compliance.
- **Cost Recovery:**
 - **Freemium model** – basic advice free, premium features paid.
 - **Subscriptions** – low-cost monthly/annual plans.

Target Market:

- Urban & rural Sri Lankan population with skin care needs.
- Public health sector & NGOs promoting affordable digital health.
- Pharmacies, insurers wanting to add digital tools.

Value Proposition:

"Equity in Dermatology"—reducing diagnostic bias while providing a 24/7 digital skin-care companion.

Feasibility:

Low-cost cloud deployment makes it scalable for NGO or government health partnerships.



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Personalized Treatment Support & Monitoring

INTRODUCTION



AI-powered skin care support system for Sri Lankan patients

Provides safe, personalized, and culturally adapted treatment guidance

Supports self-care advice, safe medicine/OTC suggestions, and progress monitoring

Multilingual system with **Sinhala and English** support

Unlike diagnosis-only tools, offers **treatment-focused and lifetime patient support**

Problems

Most existing AI-based skin care systems **focus only on diagnosis**, not treatment support.

No follow-up mechanism after initial guidance is provided.

Patients do **not receive**:

- Ongoing treatment support

- Progress monitoring

- Safety reminders or escalation alerts

Sub-Problem / Prototype Focus

Absence of **continuous treatment monitoring**

Lack of **patient engagement after first recommendation**

No system to detect **worsening conditions or red-flag symptoms**



Proposed Solution:

✓ AI-Driven Treatment & Follow-Up Support System

Provide **safe self-care advice and OTC medicine suggestions** tailored to the patient.

Introduce **notification-based follow-ups** to monitor progress:

- Day 1** – Check initial treatment response

- Day 3** – Identify improvement or worsening

- Day 7** – Decide recovery or recommend medical escalation.

OBJECTIVES

Main objective:

To develop an AI-powered system that provides **safe, personalized, and culturally adapted treatment guidance** for skin diseases in Sri Lanka.



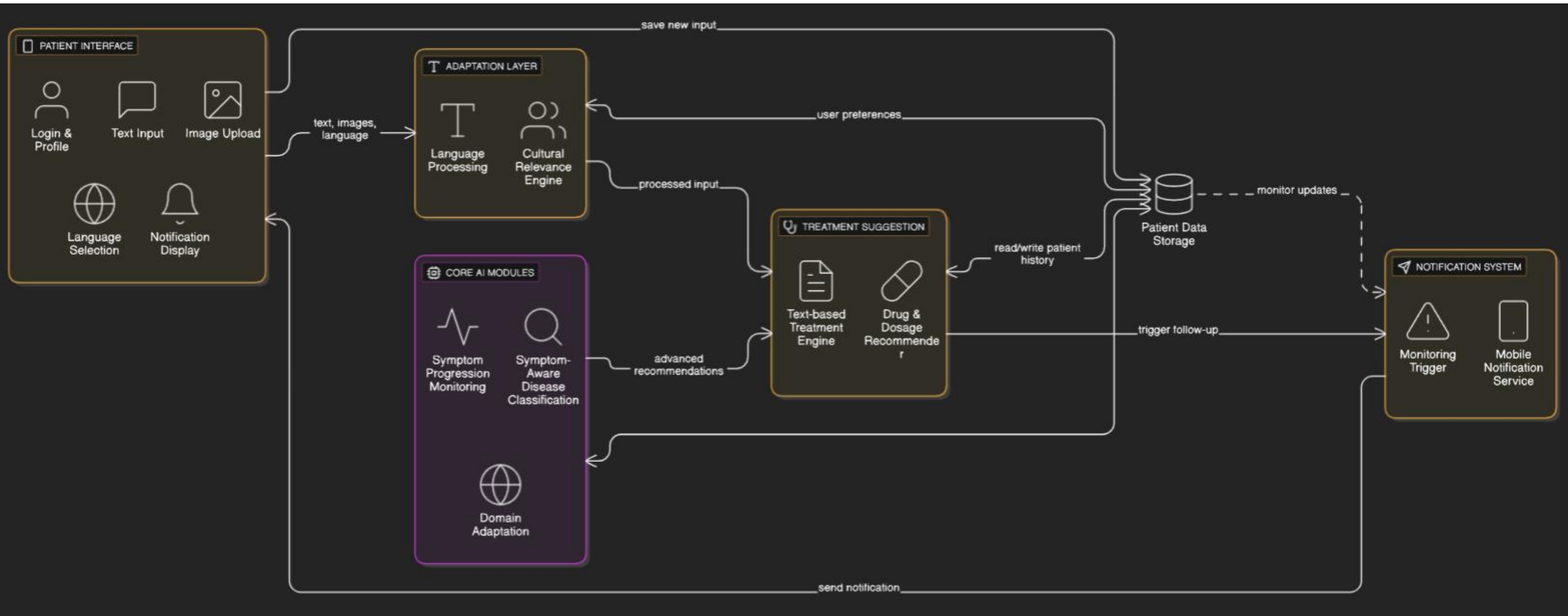
Sub Objectives:

Personalized Treatment Guidance.
Cultural and Language Adaptation.
Continuous Patient Support.
User Safety and Reliability.
Improving Access to Care



Implementation of the project

- SYSTEM OVERVIEW DIAGRAM



Functional Requirements

- Accept Sinhala/English text and optional images from patients.
- Provide treatment suggestions → self-care steps, safe OTC options, escalation rules.
- Send follow-up notifications (Day 1/3/7) and track progress.
- Maintain lifetime patient records with consent and disclaimers.

Non-Functional Requirements

- Safety & Accuracy → Red-flag detection $\geq 95\%$, treatment suggestion accuracy $\geq 85\%$.
- Performance & Reliability → Fast response ($< 3s$) and uptime $\geq 99.5\%$.
- Privacy & Security → Data encryption, access control, audit logs.
- Cultural & Language Adaptation → Clear Sinhala/English output, Sri Lanka-specific advice.

Software Requirements

- **Backend** – Python (FastAPI) / Node.js (Express) for APIs.
- **Frontend** – Web interface (React).
- **Database** – MySQL for patient data & treatment records.
- **Storage** – Cloud (for images) with encryption.
- **AI Models** – Multilingual NLP (Sinhala + English), lightweight CNN/ViT for images.
- **Notification Service** – Firebase Cloud Messaging for reminders (Day 1/3/7).
- **Security** – Data encryption, consent handling, access control, audit logs.

Personnel (Human Resource) Requirements

- **AI/ML Engineer** – Builds and fine-tunes NLP & image analysis models.
- **Backend Developer** – Implements APIs, database, and business logic.
- **Frontend Developer** – Creates patient-facing chatbot UI and notifications.
- **Database Administrator (DBA)** – Ensures secure, reliable patient record storage.

Real-World problem

Scenario

Patients develop skin conditions (rash, eczema, acne) and seek guidance online or via AI apps.
Most existing AI systems **only provide one-time diagnosis**, without treatment guidance.
No follow-up, progress monitoring, or safety alerts for worsening symptoms.
Patients may **misuse medicines, stop treatment prematurely, or delay seeing a doctor**.

Challenges

Multilingual Support: Accurate understanding of **Sinhala and English** input.

Safety & Reliability: Ensuring treatment advice is **medically safe** and **culturally appropriate**.

Patient Engagement: Encouraging adherence to notifications and treatment recommendations.

Real-World Impact

Improved Treatment Adherence: Continuous guidance ensures correct use of self-care and medications.

Early Detection of Complications: Red-flag alerts prompt timely medical intervention.

Cultural & Language Adaptation: Sinhala and English guidance increases trust and usability.

Technologies Use in the system

1. Current Technologies

Machine Learning / NLP:

TF-IDF (Term Frequency–Inverse Document Frequency) → Converts Sinhala/English text input into numerical features for model processing.

Logistic Regression → Predicts personalized treatment suggestions based on features.

Calibrated SVM (Support Vector Machine) → Improves prediction confidence and ensures safer recommendations.

Backend / Frontend Development:

VS Code → Development environment for coding and project management.

Backend (Python / Flask or FastAPI) → Handles API calls, data processing, and model inference.

Frontend (Flutter) → Builds mobile application.

Version Control & Collaboration

Git

2. Future Technologies

Machine Learning / NLP:

Multilingual Transformer Models (mBERT, XLM-R, IndicBERT) → Sinhala/English text understanding for improved accuracy.

Lightweight CNN / Vision Transformers (ViT) → Optional skin image classification for multi-modal support.

Rule-Augmented NLP Pipelines → Red-flag detection and escalation triggers for patient safety.

Knowledge Graph + LLM Integration → Personalized treatment suggestions with cultural adaptation.

Backend / Frontend / Deployment:

Notification APIs & Mobile Frameworks → Continuous patient monitoring (Day 1/3/7 follow-ups).

REST APIs (Planned) → Will connect the AI model to the frontend for real-time suggestions.



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INTRODUCTION



- **From Snapshots to Sequences**

Moves beyond single static images to track skin conditions over time.

- **Temporal Awareness**

Analyzes sequential images to detect subtle changes in symptoms day by day or week by week.

- **Progress Tracking**

Monitors improvement, stability, or worsening of skin conditions using measurable parameters.

- **Data-Driven Insights**

Quantifies changes in lesion size, color, or texture to understand disease progression.

- **Inclusive Monitoring**

- Ensures reliable tracking across diverse skin tones for fair assessment.

Problem

1. Limited to Single Images

1. Existing dermatology AI only analyzes one snapshot, missing how skin conditions evolve over time.

2. No Progress Tracking

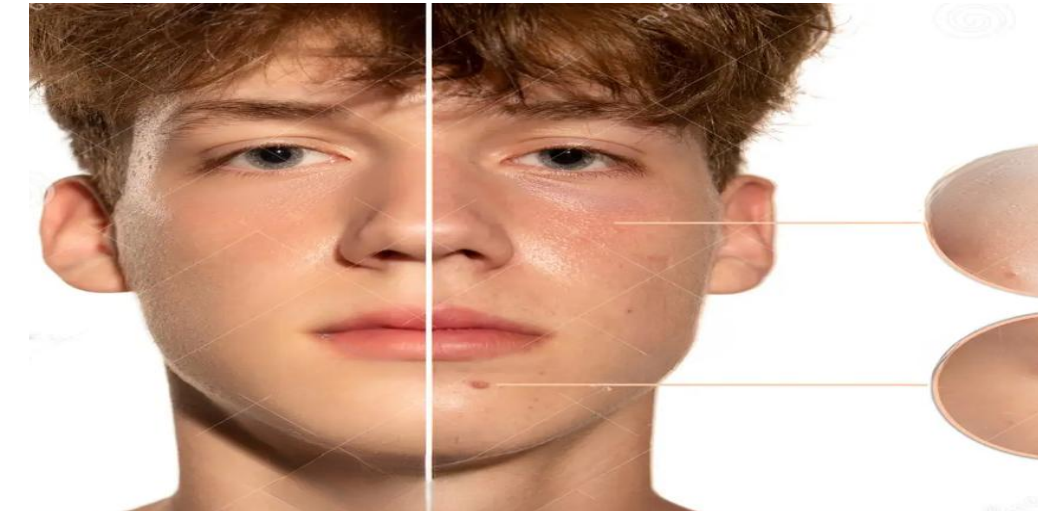
1. Patients and doctors cannot see whether symptoms are improving, stable, or worsening.

3. Subtle Changes Are Missed

1. Small changes in lesion size, color, or texture may go unnoticed without sequential analysis.

4. Bias Across Skin Tones

1. Systems may fail to accurately detect progression in darker or underrepresented skin tones.



Solution

1. Sequential Image Analysis

1. Track skin condition changes over time using daily or weekly image sequences.

2. Temporal Awareness

1. AI identifies patterns of progression, improvement, or deterioration, providing objective monitoring.

3. Parameter-Based Evaluation

1. Measures lesion size, color, and texture to quantify changes accurately.

4. Inclusive Monitoring

1. Ensures consistent and fair tracking across all skin tones, improving reliability

OBJECTIVES

Main Objective

To develop an AI-based sequential image monitoring system for tracking skin disease progression over time.



Sub-Objectives

Temporal Analysis

Analyze sequential images to detect subtle changes in skin conditions day by day or week by week.

Progress Quantification

Measure parameters like lesion size, color intensity, and texture to evaluate improvement, stability, or worsening.

Pattern Recognition

Identify trends and patterns in symptom progression for better understanding of disease dynamics.

Fair Monitoring Across Skin Tones

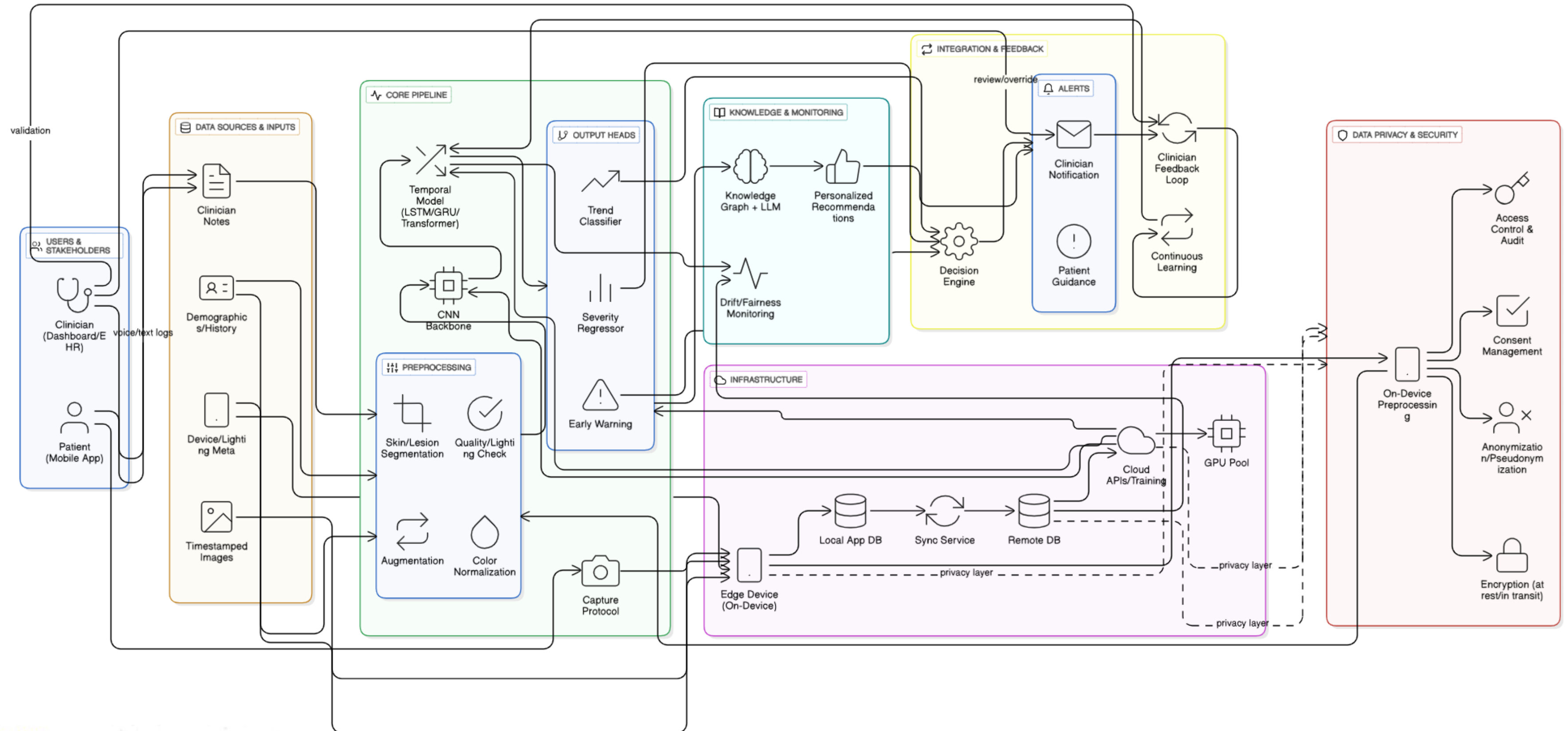
Ensure accurate tracking for all skin types, reducing bias and improving inclusivity.

Data-Driven Insights for Monitoring

Generate objective, measurable reports that visually represent progression over time.

Early Detection & Treatment Monitoring via Sequential Image Analysis

System Overview Diagram



Early Detection & Treatment Monitoring via Sequential Image Analysis

Requirements

Functional Requirements	Non-Functional Requirements
Capture sequential skin images (Day 1, Day 3, Day 7, etc.)	Accuracy $\geq 90\%$ on benchmark datasets
Preprocess inputs (segmentation, normalization, artifact removal)	Latency ≤ 5 sec on mobile devices
Temporal analysis using CNN + LSTM/GRU/Transformer	Robustness under noisy/low-quality images
Classify condition trends $\rightarrow$ <i>Improving / Stable / Worsening</i>	Fairness across diverse skin tones
Generate reports, heatmaps & alerts	Security: End-to-end encryption & HIPAA/GDPR compliance
Integration with hospital EHR & telemedicine platforms	Usability: Simple mobile interface for patients & doctors

Real world problems **Early Detection & Treatment Monitoring via Sequential Image Analysis**

Scenario

- A patient has a chronic skin condition like **eczema, psoriasis, or acne**.
- They take images at home or in clinics, but **current AI systems analyze only a single image**.
- Doctors cannot objectively see how the condition changes over days or weeks.
- Subtle worsening (increased redness, new lesions) or improvement often goes unnoticed until the next consultation.

Challenges

1.Lack of Continuous Monitoring

- 1.Existing AI cannot track changes over time, missing early signs of worsening.

2.Inconsistent Image Quality

- 1.Variations in lighting, camera angle, or resolution make sequential comparisons difficult.

3.Difficulty Detecting Subtle Changes

- 1.Small changes in lesion size, texture, or color are often invisible to both humans and static-image AI.

Impact

1.Delayed Detection of Disease Progression

- 1.Worsening symptoms may go unnoticed, leading to increased discomfort or complications.

2.Limited Patient Awareness

- 1.Patients cannot visually track how their skin condition is changing over time.



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Information Technology

Introduction

- **High prevalence of skin diseases in Sri Lanka**

Common conditions like acne, eczema, and fungal infections affect many people.

- **Limited access to dermatologists**

Rural areas often rely on general practitioners or self-medication, causing misdiagnosis and delayed treatment.

- **Early diagnosis is essential**

Current AI tools rely only on images, missing critical symptoms like itching, pain, burning, or swelling.

- **Multimodal input for better accuracy**

The system accepts **skin images** and **audio-based symptom descriptions** from users for comprehensive analysis.

- **Disease suggestion output**

After processing both visual and audio inputs, the system **classifies symptoms and suggests possible skin diseases**, supporting early and reliable diagnosis.



Problem

1. Visual similarity of skin diseases

1. Image-only classifiers (CNNs) struggle to differentiate diseases that look alike, e.g., acne vs. rosacea.

2. Text-based symptom input limitations

1. Systems requiring typing or literacy exclude rural patients who may have low literacy or are unfamiliar with typing.

3. Dataset bias

1. Most datasets are based on Western populations, leading to poor performance on Sri Lankan skin tones and culturally-specific symptom reporting.

Symptom-Aware Disease Classification



Proposed Solutions

1. Multimodal AI framework

1. Combines **CNN-based image analysis** for skin images and **RNN/speech analysis** for audio symptom descriptions.

2. Fusion layer for integrated analysis

1. Visual and symptom data are combined to improve diagnostic accuracy.

3. Natural speech input in local languages

1. Patients can speak symptoms in **Sinhala or English**, removing literacy and typing barriers.

OBJECTIVES

Main objective:

Develop a multimodal AI system for accurate skin disease diagnosis in Sri Lanka by integrating image-based and audio-based symptom inputs to support early and reliable detection, especially for underserved rural populations.



Sub Objectives:

- **Image Analysis** – Classify skin conditions with CNN; ensure accuracy on Sri Lankan skin tones.
- **Audio Symptom Analysis** – Process spoken symptoms (Sinhala/English) with RNN; detect itching, pain, burning, swelling.
- **Multimodal Fusion** – Combine image and audio data for better classification; resolve conflicts between inputs.
- **Accessibility & Usability** – Support natural speech; easy to use for rural patients.
- **Evaluation & Validation** – Test on diverse Sri Lankan datasets; outperform image-only/text-only models.

Technologies Use in the system

AI / Machine Learning-

Python Libraries: pandas for data manipulation, Pillow (PIL) for image handling, os for file path operations.

Deep Learning Framework: PyTorch (including torch, torch.nn, torch.optim, torch.utils.data, torchvision.transforms, torchvision.models) for building, training, and evaluating the deep learning model.

Machine Learning Utilities: scikit-learn (sklearn.metrics) for evaluation metrics like classification reports and confusion matrices.

Backend/Frontend

Flutter ,FastAPI ,Uvicorn

Version Control & Collaboration

Git

Functional Requirements

- Capture image + audio inputs.
- Image analysis (CNN).
- Audio analysis (RNN/Transformer).
- Fusion of multimodal features.
- Classify skin/face diseases.
- Provide results + confidence score.
- Clinician dashboard & secure storage.

Non-Functional Requirements

- Accuracy $\geq 90\%$.
- Response time ≤ 10 sec.
- Usability: Simple UI, Sinhala/Tamil voice input.
- Security: Data encryption, HIPAA/GDPR compliance.
- Scalability: Handle many users.
- Cultural adaptability & interpretability.

Software Requirements

- Languages: Python, JavaScript.
- Frameworks: TensorFlow/PyTorch, Librosa, OpenCV.
- Database: MySQL/MongoDB.
- Platforms: Web + Mobile.
- Cloud: AWS/GCP/Azure.

Personnel Requirements

- AI/ML Engineer – Image + audio model development.
- Speech Processing Specialist – Audio feature extraction.
- Dermatologist – Clinical validation & dataset labeling.
- Software Engineer – App & backend development.
- Project Manager – Coordination & integration.

Scenario:

In rural Sri Lanka, a young adult develops **facial redness, bumps, and itching**.

The nearest dermatologist is **over 50 km away**, and local clinics have only general practitioners.

The patient tries **self-medication**, which leads to **worsening of the condition**.

Challenges:

Visual similarity of skin conditions – Acne, rosacea, or fungal infections may look alike; image-only AI may misdiagnose.

Limited literacy or typing ability – Text-based symptom input is not practical for rural patients.

Cultural and skin tone differences – Most datasets are from Western populations, reducing AI accuracy on Sri Lankan skin.

Impact:

Delayed diagnosis and improper treatment.

Increased **pain, discomfort, and risk of complications**.

Healthcare inequality between urban and rural areas.

Solution via Your System:

Patient captures a **photo of the affected area** and **speaks symptoms naturally** in Sinhala/English.

The multimodal AI **analyzes image + audio** and **suggests possible skin diseases**.

Enables **early, accurate, and accessible diagnosis**, even in remote areas.

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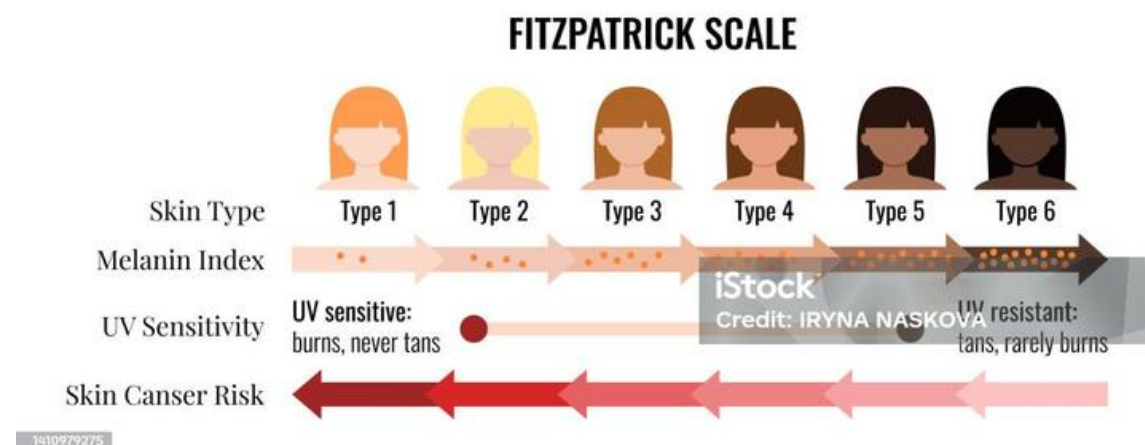


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Cross-Cultural Disease Classification via Domain Adaptation

INTRODUCTION

- Addresses **bias in skin disease AI** across light and dark skin tones
- Applies **domain adaptation** to ensure fair and accurate classification
- Includes **skin type classification** for melanin-aware diagnosis
- Integrates a **personalized skincare assistant** for individual needs
- Supports **culturally-aware healthcare** in Sri Lanka and diverse populations



Problem & Solution:

Problem

Existing skin disease AI models are **biased toward light skin tones** due to unbalanced training datasets

Poor performance on **dark and diverse skin types** leads to inaccurate or unreliable diagnoses

Lack of **skin type awareness** reduces model generalization across ethnic groups

Current systems rarely provide **personalized skincare guidance** after diagnosis

Limited support for **culturally-aware healthcare**, especially in regions like Sri Lanka

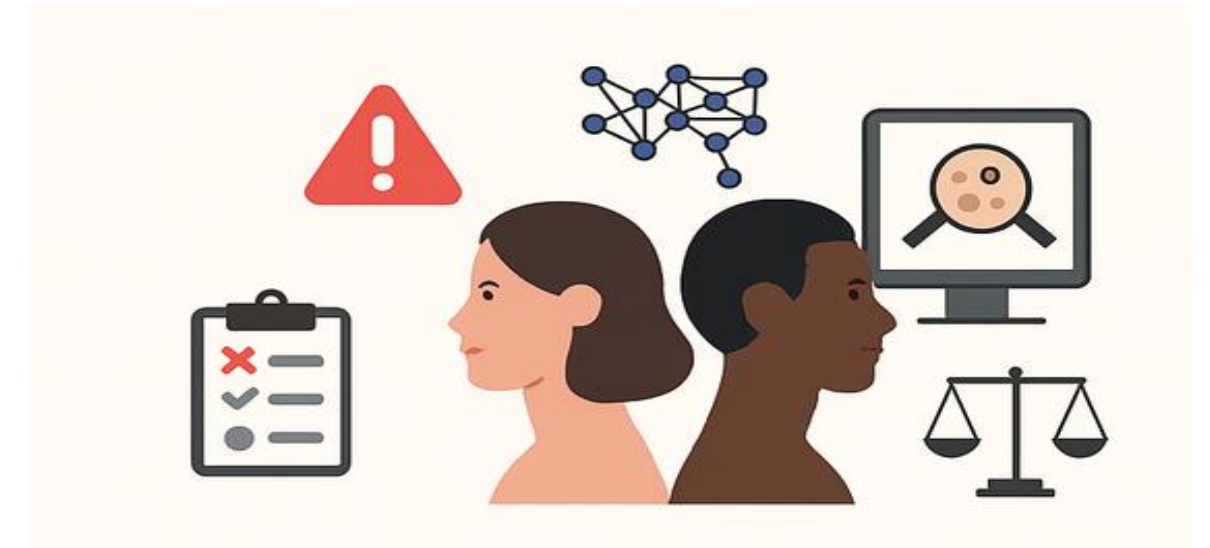
Solution

Apply **domain adaptation techniques** to reduce bias between different skin tone datasets

Integrate **skin type classification** to enable melanin-aware disease detection

Train and evaluate models using **diverse, cross-cultural datasets**

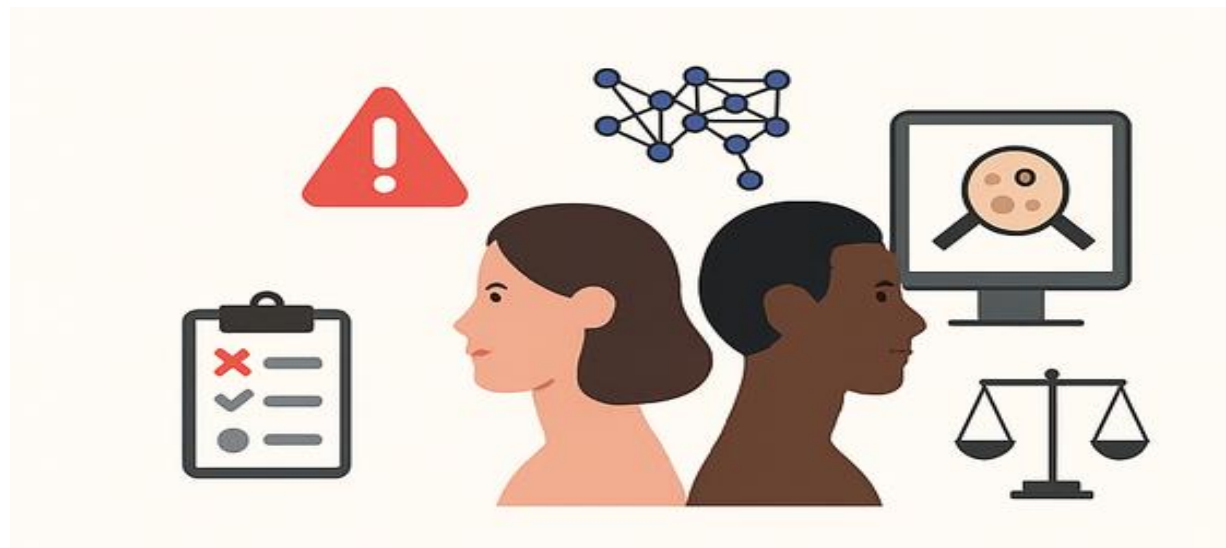
Introduce a **personal skincare assistant** tailored to individual skin type and condition



OBJECTIVES

Main Objectives

To develop a **fair and inclusive skin disease classification system** using **domain adaptation**, ensuring accurate diagnosis across diverse skin tones and cultural populations.



Sub Objectives

- Reduce **model bias** between light and dark skin tone datasets through domain adaptation techniques
- Perform **skin type classification** to enable melanin-aware disease prediction
- Improve classification accuracy for **underrepresented ethnic groups**, especially South Asian skin types
- Develop a **personalized skincare assistant** based on detected skin type and condition
- Support **culturally-aware and accessible healthcare** for Sri Lanka and global communities



Technologies Used

AI / Machine Learning-

PyTorch , timm (PyTorch Image Models) torchvision, Pillow

Model Architectures-

ConvNeXt (Final selected model) MobileNetV3 Vision Transformer (ViT) Swin Transformer

Backend/Frontend

Flutter ,FastAPI ,Uvicorn

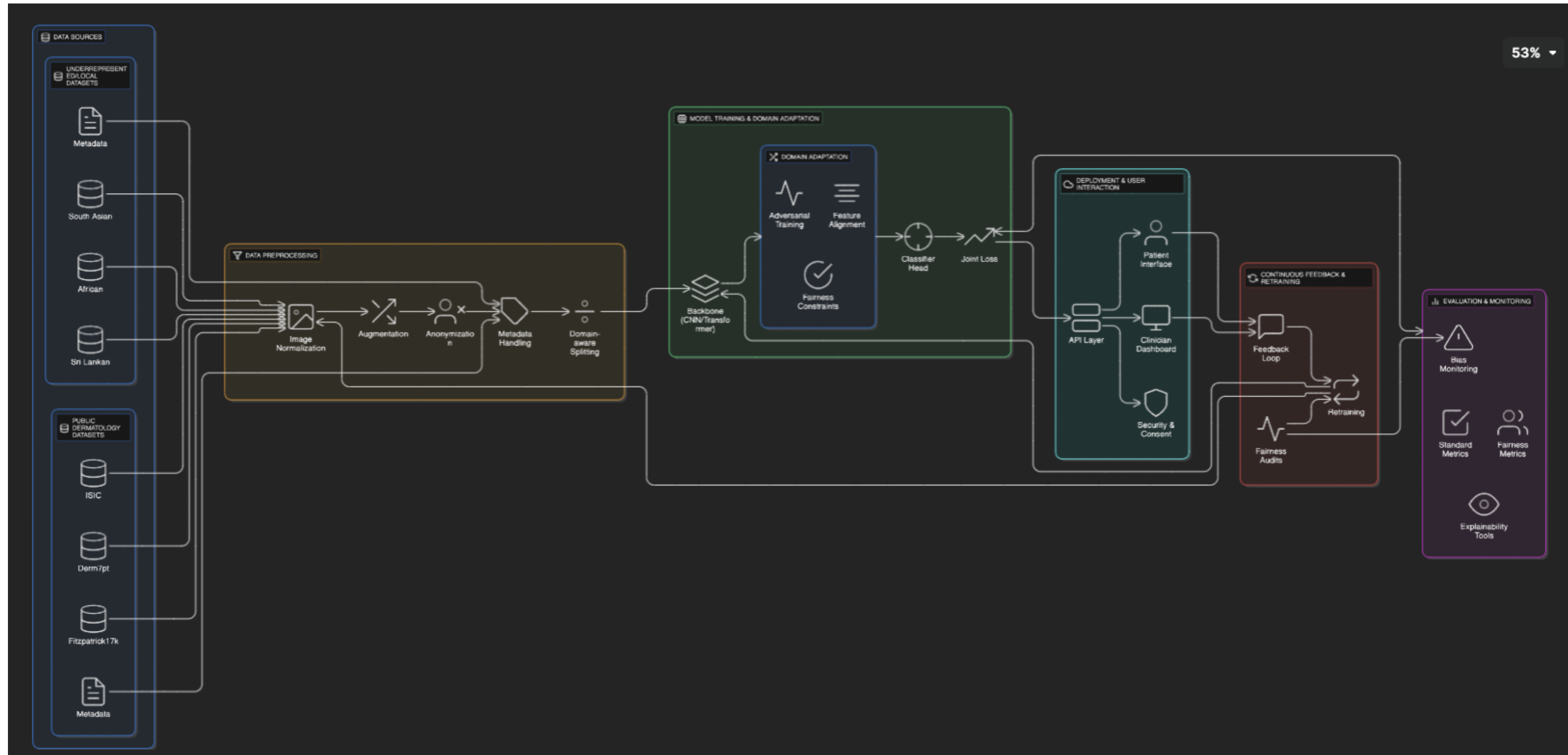
Version Control & Collaboration

Git



Implementation of the project

• SYSTEM OVERVIEW DIAGRAM



Functional Requirements

- Accept skin images from patients.
- Classify disease fairly across all skin tones.
- Provide explainable output (e.g., Grad-CAM).
- Continuous feedback loop to retrain and reduce bias.

Non-Functional Requirements

- Accuracy & Fairness: >85% accuracy, balanced error across tones.
- Performance: Fast (<3s) + reliable (uptime >99%).
- Security: Data encryption, consent handling.
- Cultural Adaptation: Clear Sinhala/English output, Sri Lanka-specific.

Software Requirements

- Backend: Python (FastAPI) for APIs.
- Frontend: Web interface (React).
- Database: PostgreSQL/MongoDB for patient + training records.
- AI Models: CNNs + Domain Adaptation (DANN, ADDA, FairDisCo).
- Explainability: Grad-CAM / SHAP.
- Security: Encryption, access control, audit logs.

Human Resource Requirements

- AI/ML Engineer: Build & fine-tune domain adaptation models.
- Backend Developer: Implement APIs + database.
- Frontend Developer: Patient-facing UI.
- Database Admin (DBA): Secure, reliable record storage.

Real World Problem

Scenario

A patient with a **dark skin tone in Sri Lanka** develops eczema.

Existing AI dermatology tools, trained mostly on light-skinned datasets, **misclassify the condition**.

Without accurate AI assistance, the patient **experiences delayed treatment**, discomfort, or worsening symptoms.

Challenge

Most AI skin disease models are trained on datasets dominated by **light-skinned individuals**.

Models **struggle to generalize** to darker skin tones due to **feature differences** (e.g., color contrast, lesion visibility).

Limited **ethnic and regional data** (e.g., South Asian, Sri Lankan skin types) hinders model reliability.

Incorporating **skin type variations** and **cultural skincare practices** is complex for AI systems.

Impact

Misdiagnosis or delayed diagnosis for people with darker skin tones.

Reduced **trust in AI healthcare solutions**, potentially widening healthcare inequality.

Patients may **receive incorrect treatment or skincare advice**, affecting outcomes.

Overall, healthcare systems fail to deliver **fair and inclusive dermatological support**.

Thank You